



Lab Title: IP Address Lab Exercise

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Executive Summary

This lab exercise focuses on understanding and applying core concepts of **IP addressing and subnetting** in computer networks. It introduces the foundational principles behind IPv4 structure, address classification, and routing. Through guided questions and problem-solving, the exercise helps build technical awareness of how devices communicate within and across networks.

Lab Objectives

The main objectives of this lab are to:

1. Understand the difference between **flat and hierarchical addressing** systems.
2. Explain the roles of **IANA and RIRs** in global IP management.
3. Identify and classify **IP address classes (A, B, C)** and determine their default subnet masks.
4. Differentiate between **static and dynamic addressing** and explain when each is used.
5. Perform **network, broadcast, and host address calculations** for given IPs.
6. Apply **subnetting techniques** to calculate usable host ranges and improve IP efficiency.
7. Recognize **special-purpose addresses** such as loopback, APIPA, and multicast.
8. Understand the function of **default gateways** and direct vs. indirect delivery.
9. Analyze **common configuration errors** and explain how they affect connectivity.

Tools & Resources

- I used **Kali Linux** to run basic network commands and check configurations.
- Commands like **ping** and **ipconfig** helped me test and understand how devices communicate.
- I also used an **online subnet calculator** to double-check my IP and subnet results.
- For learning support, I watched **YouTube tutorials** and reviewed the **ICDFA lab guide** and **Cisco networking materials** to better understand how IP addressing works.

Methodology

❖ Preparation and Setup

- ✓ Configured virtual network devices (e.g., PCs, routers, switches) to test addressing schemes.
- ✓ Verified connectivity tools such as `ping` and `ipconfig` were working.

❖ Section A - Core Concepts

- ✓ Defined key networking terms such as *Flat vs. Hierarchical Addressing*, *Static vs. Dynamic IP*, etc.
- ✓ Used textbook and online resources to understand differences between each concept.

❖ Section B - Address Analysis

- ✓ Identified the IP class (A, B, or C) of given addresses.
- ✓ Calculated **network address**, **broadcast address**, and **default subnet mask** for each IP.
- ✓ Verified results using subnet calculators and manual binary calculations.

❖ Section C - Calculations and Problem Solving

- ✓ Solved practical subnetting problems (e.g., /27, /24).
- ✓ Calculated total and usable host addresses.
- ✓ Determined **default gateway** relationships and reasons for connectivity issues.
- ✓ Practiced identifying **special-purpose addresses** (e.g., loopback 127.0.0.1 and APIPA 169.254.x.x).

❖ Section D - Challenge Scenario

- ✓ Analyzed network misconfigurations in a real-world scenario.
- ✓ Used logical reasoning to identify incorrect subnet mask or gateway settings.

Section A: Core Concepts

Fig 1. Flat vs. Hierarchical Addressing

Flat Addressing: The address is a unique identifier, but it contains **no inherent information** about the item within the system. It's like a random Social Security Number.

Hierarchical Addressing: The address is structured into multiple parts, where **each part specifies a location within a nested hierarchy**. It's like a postal address.

FEATURES	FLAT ADDRESSING	HIERARCHICAL ADDRESSING
Structure	No logical structure	Divided into network and host parts
Scalability	Poor for large networks	Highly scalable
Example	MAC Address	IP Address
Routing Efficiency	Inefficient	Efficient
Use Case	Small local network	Internet and enterprise
Analogy	Asocial security number (unique to you, but doesn't say where you live)	A postal address (Country -> State -> City -> Street -> House Number).
Core Principle	Address is a simple, unique identifier	Address is structured, defining a path through a hierarchy.

Fig 2. IANA and RIRs

IANA (Internet Assigned Number Authority)

The Internet Assigned Number Authority is the World Library Headquarters. It holds the master catalog of all available books (IP address) and allocates entire encyclopedia sets (large blocks of IP addresses, DNS root zones and protocol assignments) to continental libraries, to ensure the smooth operation of global internet.

IP Address Allocation: IANA holds the global pool of all **IPv4** and **IPv6** addresses. It allocates huge, contiguous blocks of these addresses to the five **RIRs**

Protocol Number Management: it manages the codes and the numbers that makes internet protocol work, such as port numbers (e.g., **Port 80** for **HTTPs**), for protocol number, and enterprise numbers.

DNS Root Zone Management: IANA oversees the top-level of the Domain Name System (**DNS**), managing the list of top-level domains (**TLDs**) like .com, .org, .uk, and country codes.

Function: IANA manages the root of the internet's addressing system. It doesn't assign addresses to individual companies or people but to the large regional bodies below it. The IANA functions are performed by the **Internet Corporation for Assigned Names and Numbers (ICANN)**

Fig 2.1. RIRs (Regional Internet Registry)

RIRs are the **Continental Libraries** (e.g., the North America Library, the European Library). They receive large blocks of IP address allocation from **IANA** and are responsible for distributing them to **Internet Service Providers (ISPs)**, **government**, and **organizations** within their regions.

Allocating IP Resources to LIRs: They provide smaller blocks of **IPv4** and **IPv6** address to **Local Internet Registries (LIRs)**, which are almost always ISPs

Assigning ASNs: they assign **Autonomous System Numbers (ASNs)**, which are unique identifiers for large networks (like an ISP's network) on the internet.

Registration: They maintain the public WHOIS database for their region, which records who holds which IP addresses and ASNs. This is critical for troubleshooting and networking operations.

Public Development: They facilitate a community-driven process where members (the ISPs and organizations) create policies for how resources should be distributed fairly within that region.

Fig 3. Requirements for Internet Routing

- i. IP Addressing**
 - Every device needs a unique IP address to send and receive data.
 - Proper subnetting helps in efficient route management.
- ii. Routing Protocols**
 - Routers use routing protocols to exchange information and determine best paths:
 - Interior Gateway Protocols (IGPs) – e.g., RIP, OSPF, EIGRP
 - Exterior Gateway Protocols (EGPs) – e.g., BGP (used on the internet)
- iii. Routing Table**
 - Maintains a list of known networks and how to reach them.
 - Built dynamically (via protocols) or statically (manually set).
- iv. Routers**
 - Physical or virtual devices that forward data between networks based on IP.
- v. Network Topology**
 - A clear and efficient design of how networks and routers are interconnected.
- vi. Autonomous Systems (AS)**
 - Groups of IP networks managed by one organization, identified by an AS number.
 - Important for global internet routing (BGP).
- vii. Path Determination**
 - Routers choose the best path using metrics (e.g., hop count, bandwidth, and delay).
- viii. Forwarding**
 - Routers must be able to forward packets correctly based on the destination IP.

Fig 4. Static vs. Dynamic IP Address

Static IP Address

An IP address that is permanent, manually assigned to a device that does not change. Hosting a website or server where the IP must remain constant (e.g., A company's website, www.google.com). Uses a static IP so users can always reach it.

Dynamic IP Address

An IP address that is temporary assigned automatically by a **Dynamic Host Configuration Protocol (DHCP)** server and may change over time. Most home internet connections, where fixed IP isn't needed (e.g., regular browsing).

FEATURE	Static IP Address	Dynamic IP Address
Definition	An IP address that is manually assigned to a device and does not change over time.	An IP address that is automatically assigned by a DHCP server and changes periodically.
Assignment Method	Manually configured by an administrator.	Automatically assigned by DHCP (Dynamic Host Configuration Protocol).
Stability	Remains constant unless manually changed.	Can change each time the device connects to the network.
Cost	Usually more expensive to maintain.	Cheaper and easier to manage.
Security	More vulnerable to hacking since it's predictable.	More secure since it changes over time.
Management	Requires more administrative effort.	Easier to manage, especially for large networks.
Best For	Servers, websites, and network devices that need constant access.	Personal devices like laptops, phones, and home routers.

Fig 5. Classful vs. Classless Addressing

1. **Classful Addressing** divides IP addresses into fixed classes (**A, B, C, D, and E**) with predefined subnet masks. It wastes IP addresses because the ranges are rigid.
2. **Classless Addressing (CIDR - Classless Inter-Domain Routing)** allows flexible allocation of IP addresses using custom subnet masks (e.g., /22, /26), making it more efficient.

Advantage of Classless Addressing:

- Better IP utilization - It reduces waste by allowing networks to be assigned only the number of addresses they actually need.

Section B: Address Analysis

IP Address	Class	Network Address	Broadcast Address	Default Subnet Mask	Points
10.50.100.10	A	10.0.0.0	10.255.255.255	255.0.0.0	2
172.16.0.25	B	172.16.0.0	172.16.255.255	255.255.0.0	2
192.168.1.150	C	192.168.1.0	192.168.1.255	255.255.255.0	2
91.118.125.2	A	91.0.0.0	91.255.255.255	255.0.0.0	2
203.0.113.45	C	203.0.113.0	203.0.113.255	255.255.255.0	2

Section C: Calculations & Problem Solving

Fig 16. Special Purpose Address: refers to a group of IP addresses that are reserved for specific network functions rather than normal device-to-device communication over the internet.

The IP address **127.0.0.1** is known as the **loopback address**.

It is used to **test network connectivity within a local computer** - meaning any data sent to this address stays inside the same device.

Fig 17. APIPA Address

An IP address like 169.254.10.1/16 indicates that the device has been assigned an APIPA (Automatic Private IP Addressing) address.

It means the system is trying to self-configure but can't connect to the wider network or internet. This often points to a DHCP failure or network issue.

Fig 18. Default Gateway

Why must the default gateway be on the same network as the host device?

A device can only directly communicate with other devices on the same network.

If the gateway (usually a router) is on a different network, the host won't be able to reach it to forward traffic outside its local network - meaning internet or external access won't work.

In simple terms: It's like trying to talk to a neighbor using a walkie-talkie - you must be on the same channel (network) to connect.

Fig 19. Usable Host Calculation

For network 192.168.50.0/24, calculate:

- ✓ Total possible host addresses: _____
- ✓ Number of usable host addresses: _____

Network: 192.168.50.0/24

the prefix “/24” indicates that 24 bits are used for the network portion, IP addresses are 32 bits total. So, **32 - 24 = 8 bits**, Leaving 8 bits for host addressing.

Step 1: Calculate total possible host addresses

$$2^8 = 256 \text{ total addresses}$$

Step 2: Subtract reserved addresses

- **1 address** for the network (192.168.50.0)
- **1 address** for broadcast (192.168.50.255)

$$256 - 2 = 254 \text{ usable host addresses}$$

✓ **Final Answer:**

- **Total possible host addresses:** 256
- **Usable host addresses:** 254

This means the 192.168.50.0/24 network can accommodate up to **254 active**

Fig 20. Classless Addressing

For 198.51.100.0/27:

- a) Subnet mask in dotted-decimal: _____
- b) Number of usable hosts: _____

a) Subnet mask in dotted-decimal:

A /27 prefix means 27 bits are for the network.

So, subnet mask in binary:

11111111.11111111.11111111.11100000

Convert to dotted-decimal:

255.255.255.224

b) Number of usable hosts:

$2^5 (32 - 27) = 2^5 = 32$ total addresses

Usable = $32 - 2$ (network + broadcast) = 30

Final Answers:

a) Subnet mask: 255.255.255.224

b) Usable hosts: 30

Fig 21. Network Design Efficiency

Your boss wants to use 192.168.10.0/24 for 12 computers. Is this efficient? Why or why not?

No, it's not efficient.

Using 192.168.10.0/24 gives you 254 usable IP addresses, but you only need 12 for the computers. This means most of the IPs will be unused, wasting address space.

Better Option:

Use a smaller subnet like /28, which gives 14 usable IPs just right for 12 computers plus a few extras (e.g., router, printer).

Summary:

/24 = 254 usable IPs → too much for 12 devices

/28 = 14 usable IPs → more efficient and appropriate

Fig 22.Multicast Address Recognition

True or False: 224.0.0.10 can be assigned to a laptop. Explain.

False.

224.0.0.10 is a multicast IP address, not meant for assignment to individual devices like a laptop.

Explanation:

IPs in the 224.0.0.0 – 239.255.255.255 range are reserved for multicast traffic.

Multicast is used to send data to multiple devices simultaneously (e.g., streaming or routing protocols).

Laptops and other end-user devices should be assigned unicast addresses, typically in the private IP range (like 192.168.x.x).

Conclusion: Multicast addresses like 224.0.0.10 are not for direct assignment to devices.

Fig 23. Direct vs. Indirect Delivery

Device 10.10.10.10/24 needs to send data to 10.10.11.10. Will it use the gateway? Why?

Yes, it will use the gateway.

Why:

Device IP: 10.10.10.10/24 -> subnet mask 255.255.255.0

Destination IP: 10.10.11.10

With a /24 subnet, only addresses from 10.10.10.0 to 10.10.10.255 are in the same network.

Since 10.10.11.10 is outside this range, it's not on the same network, so the device must send the data to the default gateway for indirect delivery.

The gateway will then forward the packet toward the correct destination network.

Section D: Challenge Scenario

Fig 24.Troubleshooting Network Configuration

A computer has the following configuration:

- ✓ IP: 172.20.150.5
- ✓ Subnet Mask: 255.255.0.0
- ✓ Default Gateway: 172.21.150.1

The user cannot access the internet. Based on the IP addressing rules you've learned, what is the most likely configuration error?

The most likely configuration error is:

✗ The Default Gateway is not in the same network as the computer.

Here's why:

IP address: 172.20.150.5

Subnet mask: 255.255.0.0 -> Network range: 172.20.0.0 to 172.20.255.255

Default Gateway: 172.21.150.1 -> this is outside the computer's network.

Fix:

The default gateway should be something within the 172.20.0.0/16 network, like:

✓ 172.20.150.1

The device can't communicate with a gateway that's not in its own subnet.

Recommendations

1. **Ensure Proper Network Planning:**
Before assigning IP addresses, perform subnet planning to avoid IP conflicts and optimize address usage.
2. **Adopt Classless Addressing (CIDR):**
Use classless addressing to improve IP address efficiency and scalability, especially for large or growing networks.
3. **Implement Centralized IP Management:**
Organizations should use IP management tools (such as DHCP servers) to automate IP allocation and reduce manual errors.
4. **Maintain Accurate Documentation:**
All assigned IP addresses, subnet ranges, and gateways should be well documented for easier troubleshooting and auditing.
5. **Regularly Review Network Configuration:**
Periodic checks and network audits should be carried out to ensure proper configuration, connectivity, and compliance with best practices.

References

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